

NFTS HOST BASED USB Device Forensic Analysis Report

CASE TYPE	USB FORENSICS
DATE OF REPORT	July 2026
Analyst	Julian Derry
REPORT STATUS	COMPLETED

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Executive Summary

This report documents a digital forensic examination of a Windows 11 virtual machine (win_11_vm) to identify and analyze USB device activity on the system. The examination focused on confirming the identity of a connected USB device, determining the assigned drive letter, identifying the Windows user who interacted with the device, identifying files accessed from the device, and determining whether files were copied between the USB device and the host computer. Registry artifacts, file system artifacts, and application artifacts were correlated to build a timeline of USB activity.

Scope and Objectives

The examination sought to answer the following questions:

- What is the identity of the USB device connected to the system, including manufacturer, model, and serial number?
- What drive letter was assigned to the USB device?
- Which files were opened from the USB device?
- Which Windows user profile interacted with the USB device?
- Which applications opened files from the USB device?
- Were any files copied between the USB device and the computer?

Host Details

Attribute	Value
Device name	win_11_vm
Processor	13th Gen Intel(R) Core(TM) i5 13420H (2.61 GHz) (2 processors)
Installed RAM	8.00 GB
Graphics card	VMware SVGA 3D (4 MB)
Storage	37 GB of 64 GB used
System type	64 bit operating system, x64 based processor
Pen and touch	Touch support with 8 touch points
Edition	Windows 11 Pro
Version	25H2
Installed on	6/25/2026
System Timezone	GMT
OS build	26200.8875
Experience	Windows Feature Experience Pack 1000.26100.334.0

Find a setting

System > About

Processor

**13th Gen Intel(R)
Core(TM) i5-13420H**

2.61 GHz (2 processors)

Installed RAM

8.00 GB

DRAM

Graphics card

4 MB

VMware SVGA 3D

Storage

64 GB

37 GB of 64 GB used

win_11_vm
VMware20,1

Rename this PC



Device info

Copy



Device name	win_11_vm
Processor	13th Gen Intel(R) Core(TM) i5-13420H (2.61 GHz) (2 processors)
Installed RAM	8.00 GB
Graphics card	VMware SVGA 3D (4 MB)
Storage	37 GB of 64 GB used
Device ID	
Product ID	
System type	64-bit operating system, x64-based processor
Pen and touch	Touch support with 8 touch points

Related links [Domain or workgroup](#) [System protection](#) [Advanced system settings](#)



Windows info

Copy

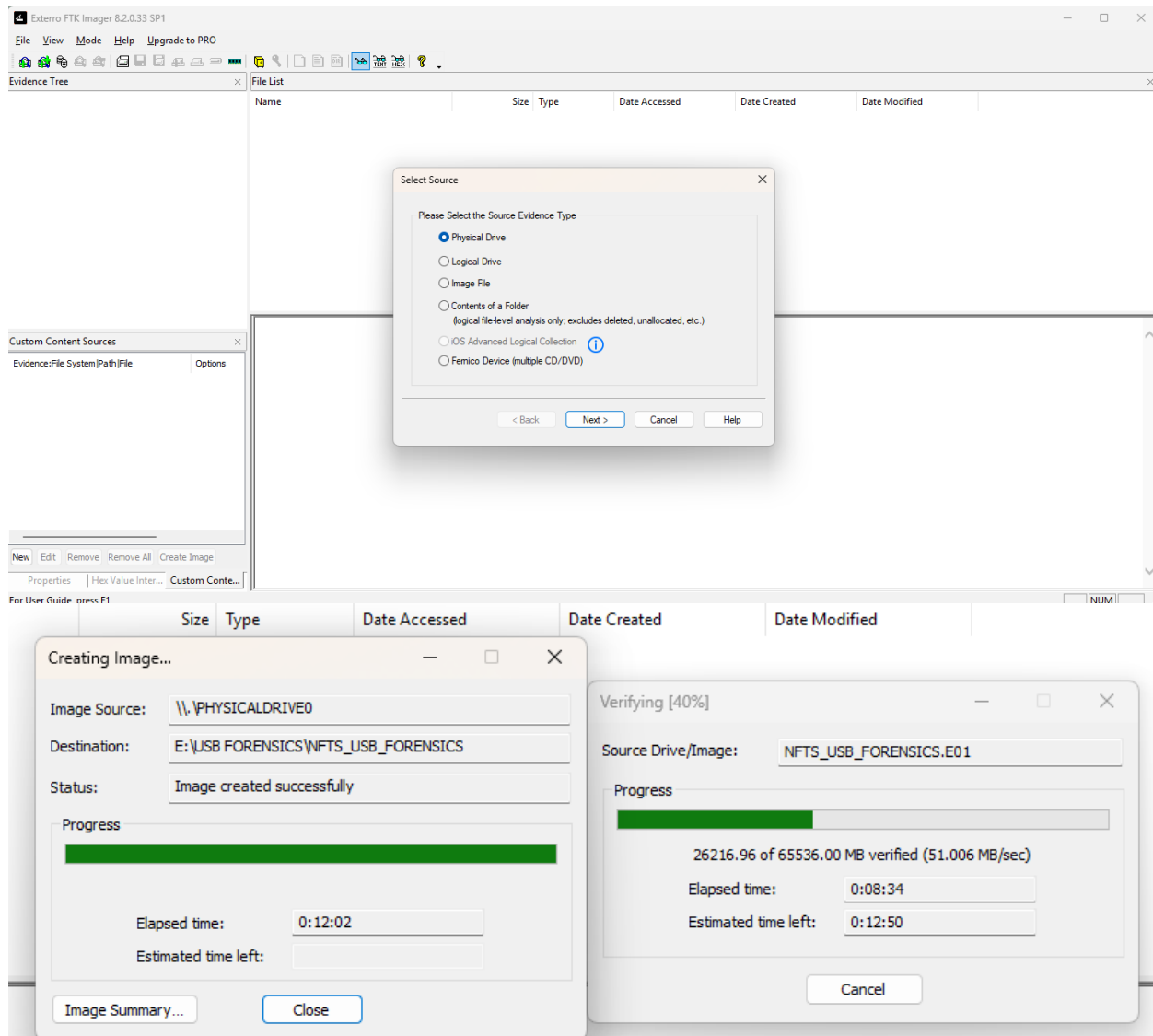


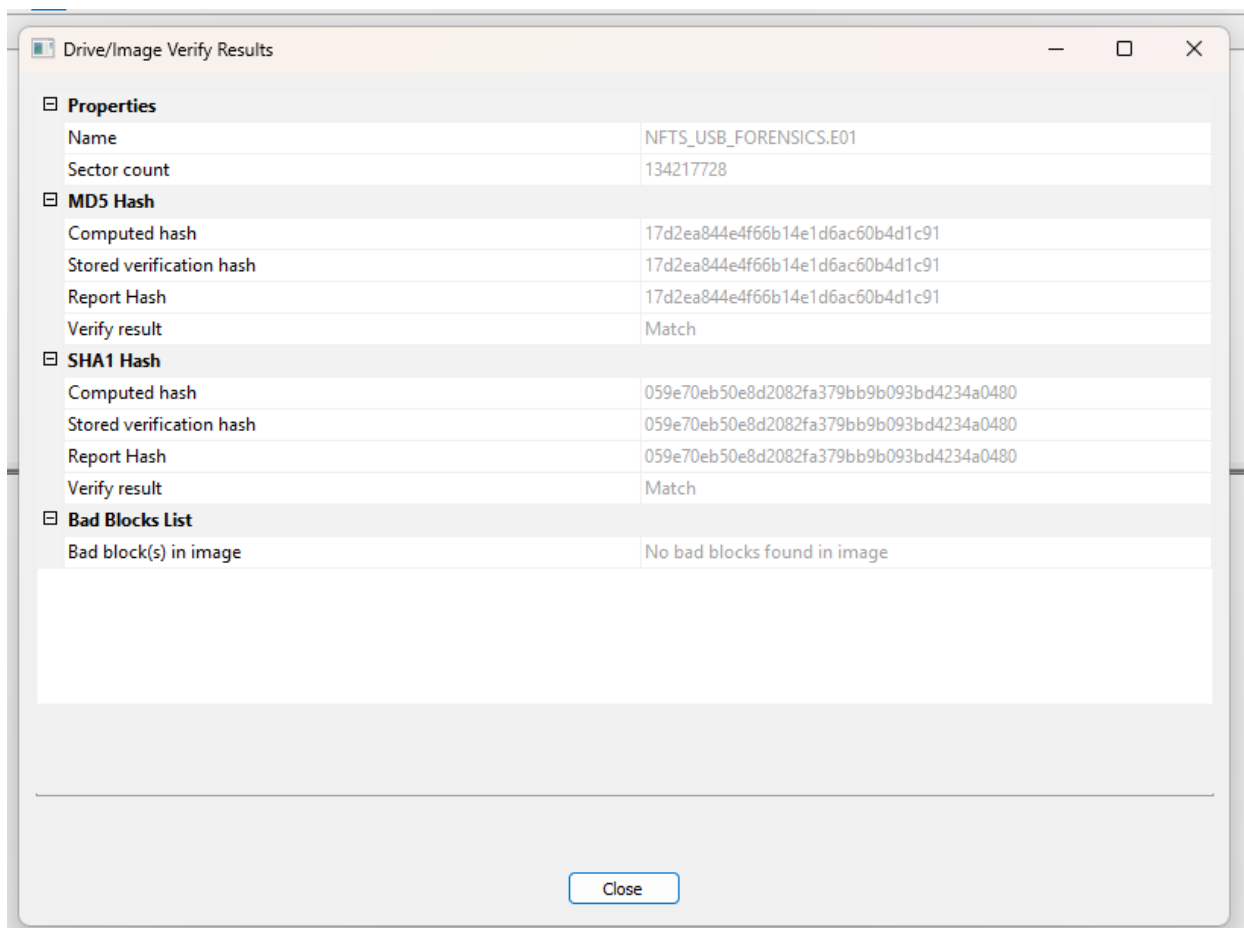
Edition	Windows 11 Pro
Version	25H2
Installed on	6/25/2026
OS build	26200.8875

Evidence and Chain of Custody

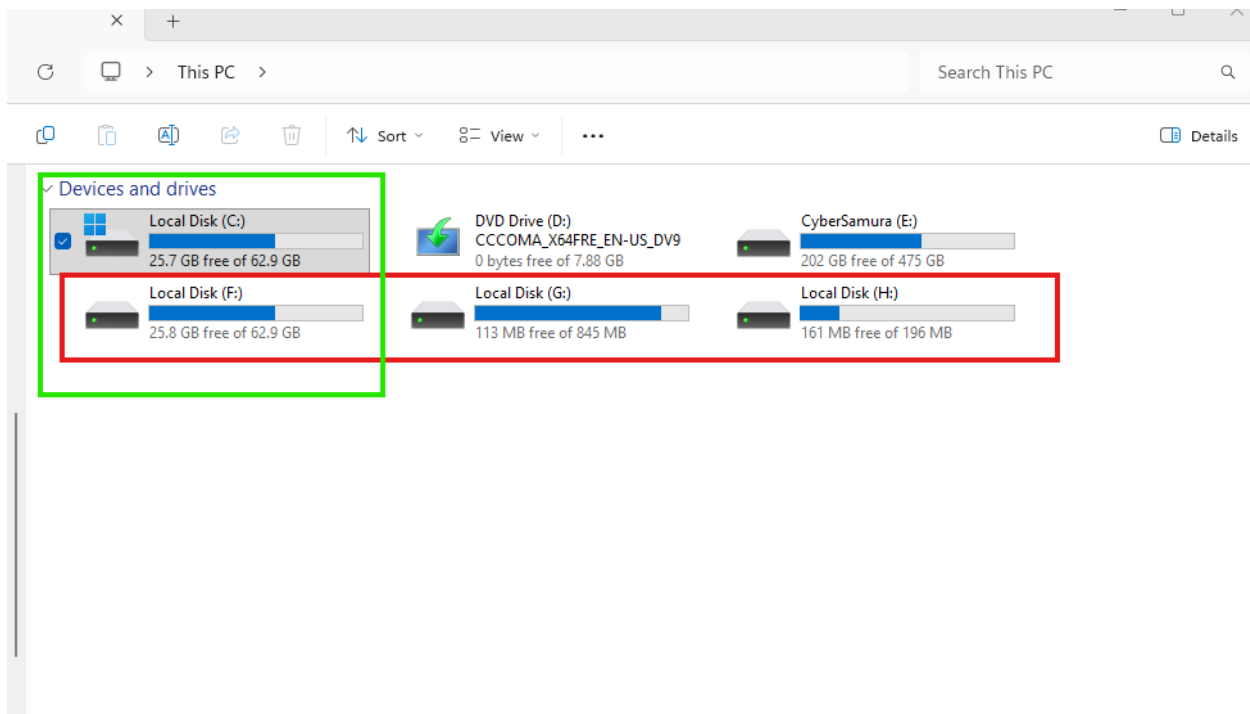
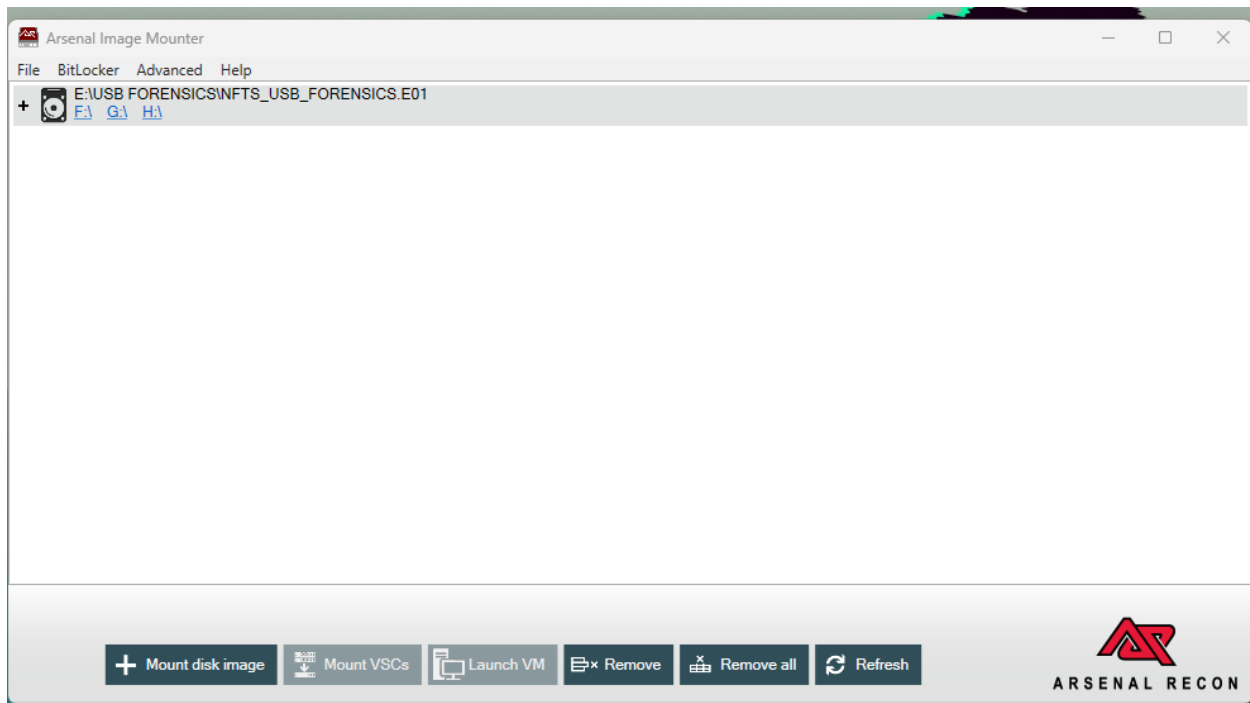
An E01 image was acquired, and hash values were calculated and verified.

Attribute	Value
Image file name	NFTS_USB_FORENSICS.E01
Image Name	NFTS_USB_FORENSICS.E01
Acquisition Tool	FTK Imager v8.2.0.33 SP1
Acquisition Date	7/20/2026
MD5 Hash	17d2ea844e4f66b14e1d6ac60b4d1c91
SHA1 Hash	059e70eb50e8d2082fa379bb9b093bd4234a0480
Examiner	Julian Derry





The image was mounted using Arsenal Image Mounter v3.12.331. This approach was taken because mounting the image as a physical disk, rather than only a logical volume, allowed other tools used in this examination, including KAPE, Eric Zimmerman's tools, and virtual machine software, to interact with it as actual storage. The image was mounted as a Disk device with write temporary changes stored in an alternate differencing file location, so as not to modify or alter the original image file. The mounted image was assigned drive letter F.



Tools and Methodology

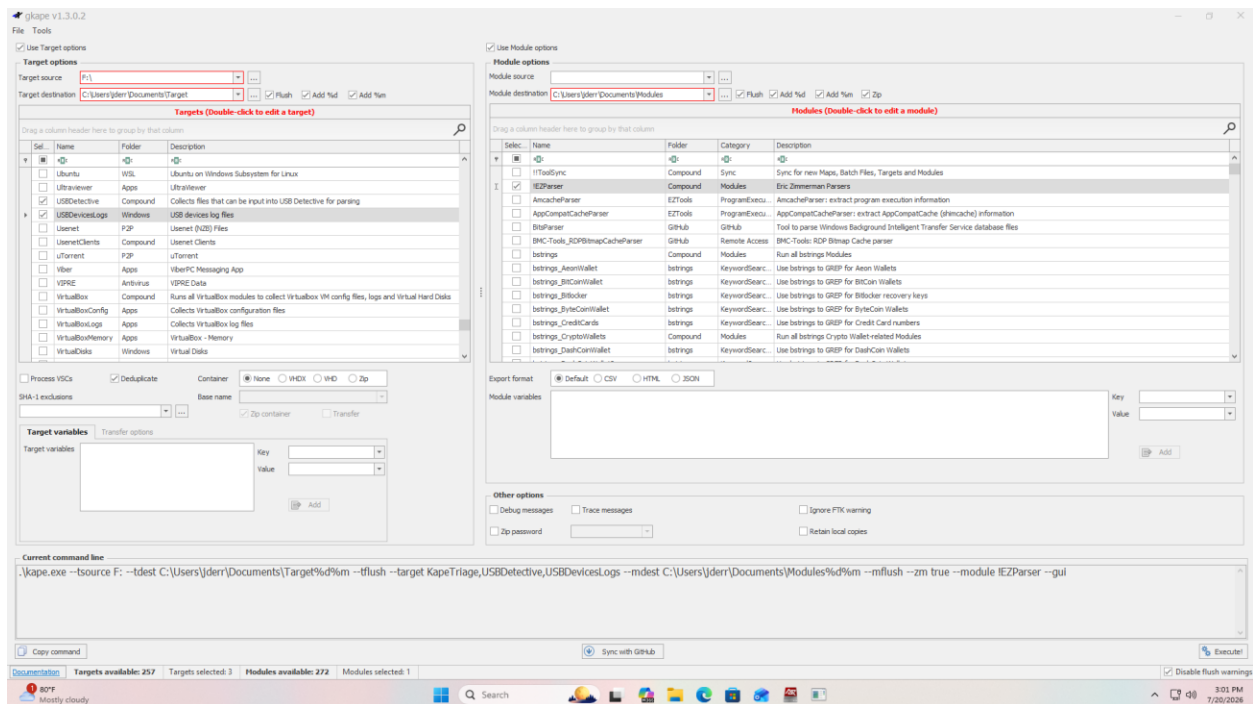
The following tools were used during this examination:

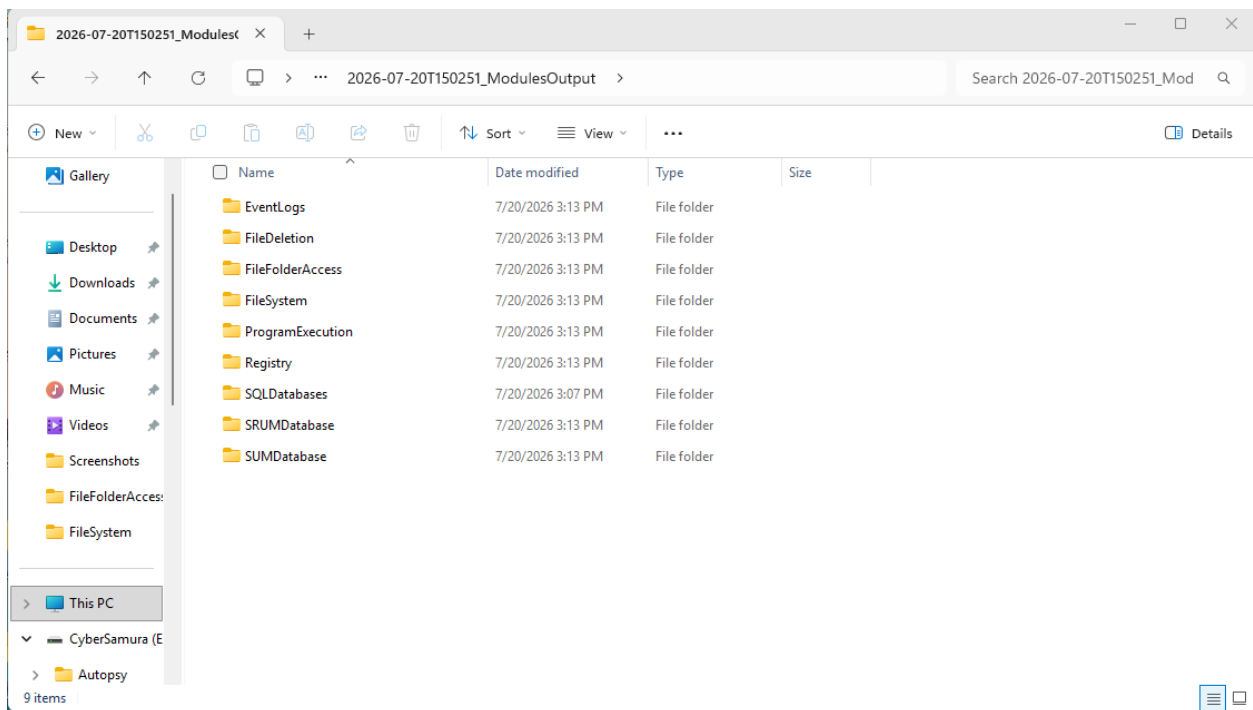
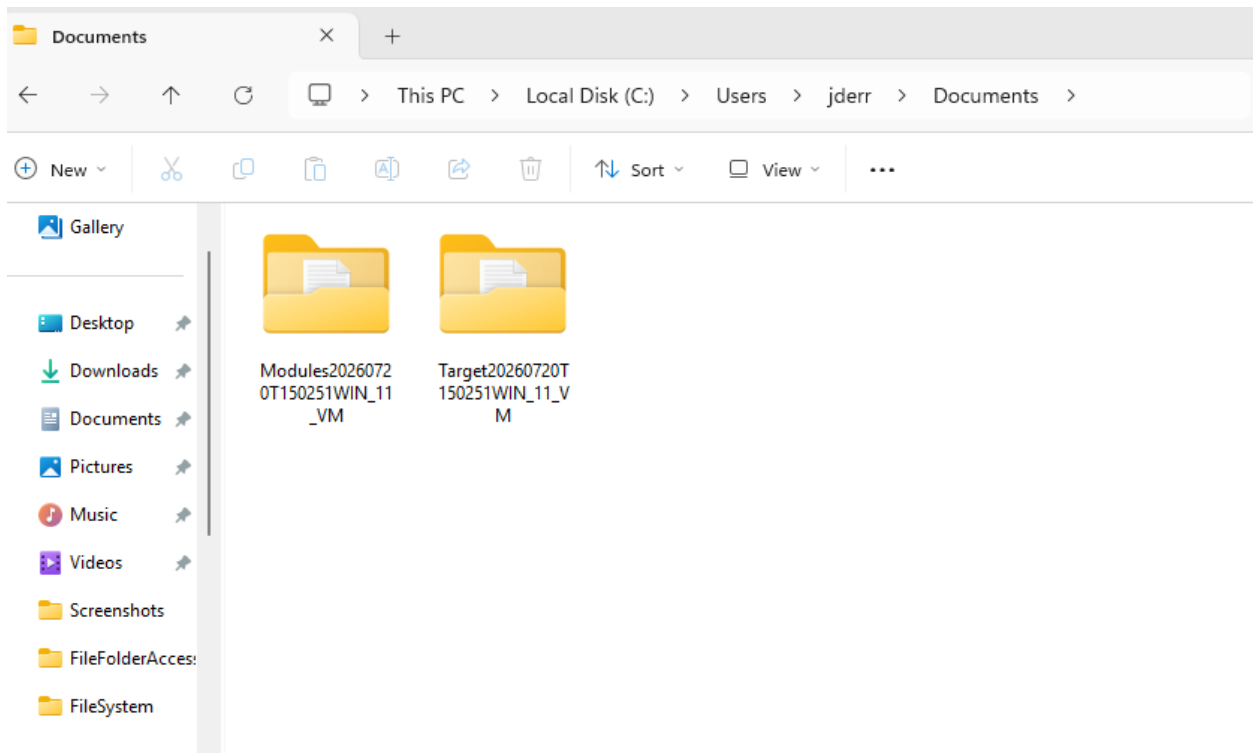
- FTK Imager v8.2.0.33 SP1, for acquisition
- Arsenal Image Mounter v3.12.331, for mounting the E01 image

- KAPE (gkape v1.3.0.2), for targeted collection and processing
- EZParser module, run within KAPE
- MFTECmd, RECcmd, and EvtxECmd, run as KAPE modules for timeline creation
- Registry Explorer, for registry hive analysis
- USB Detective, for USB artifact verification
- Timeline Explorer, for reviewing KAPE module output CSV files

KAPE was run to collect Registry targets, specifically NTUSER.DAT, SYSTEM, SOFTWARE, SAM, SECURITY, and USRCLASS.DAT, along with the USBDetective target and supporting artifacts including \$J, \$LOGFILE, and \$MFT. The USBDetective Windows application was also used separately for redundancy, along with USBDeviceLogs, to provide redundancy for USB artifacts and logs.

For timeline creation, the following KAPE modules were run: EZParser, MFTECmd, RECcmd, and EvtxECmd. The module output folder contained several folders of categorized CSV files, which were reviewed using Timeline Explorer.





Findings

USB Device Identity

Registry Explorer was used to load the SYSTEM hive and examine SYSTEM\CurrentControlSet\Enum\USBSTOR. Three USB devices were present under this key. The examination focused on the most recently inserted device, connected on 2026 07 20 at 13:23:17. Manufacturer, version, serial number, device name, installed, first installed, last connected, and last removed data were available under this key.

Timestamp	Manufacturer	Title	Version	Serial Number	Device Name	Disk ID	Installed	First Installed	Last Connected	Last Removed
2026-07-20 13:25:27	Ver_Flash	Prod_USB_Special_Disk	Rev_...	8&25cc1fb8&0	Flash USB Special Disk	USB Device	2026-07-20 13:25:27	2026-07-20 13:25:27	2026-07-20 13:26:55	2026-07-20 13:27:56
2026-07-20 13:23:17	Ver_Server	Prod_Mass Storage	Rev_2.00	202404180000000	Generic Mass Storage	USB Device	2026-07-20 13:23:17	2026-07-20 13:23:17	2026-07-20 13:23:17	2026-07-20 13:23:38
2026-06-25 10:25:17	Ver_VendorCo	Prod_ProductCode	Rev_2.00	5280311417938667	VendorCo ProductCode	USB Device	2026-06-25 10:25:17	2026-06-25 10:25:17	2026-07-20 13:25:21	2026-07-20 13:31:21

The value 8&25cc1fb8&0 seen under USBSTOR is a Windows generated ParentIdPrefix, created when a USB device does not report a unique serial number to the operating system. This can be identified by the presence of ampersands in the value.

To verify the actual device serial number, the following locations were checked:

- SYSTEM\CurrentControlSet\Enum\USB
- MountedDevices
- SetupAPI.dev.log

SetupAPI.dev.log may explicitly indicate that a device has no serial number, or may show how Windows enumerated the device. In this case, SYSTEM\CurrentControlSet\Enum\USB showed the same date and time as USBSTOR, but with a serial number of 20240418000000, without the ampersands present in the USBSTOR entry. USB Detective was also used for verification and returned a device with the same serial number, 20240418000000, and a similar timeline.

Registry Explorer v2026.5.0

File Tools Options Bookmarks (36/0) View Help

Registry hives (2) Available bookmarks (72/0)

Enter text to search...

Find

Key name

C:\Users\juser\Documents\Target20267207150251WRL_11_V

ROOT

ActivationDriver

ControlSet001

Control

Enum

ACPI

ACPI_HAL

PCIDE

ROOT

CDisplay

HEAUIED

MSD

HTREE

ACI

PCIDE

ROOT

SCSI

STORAGE

SWIO

SWIO

UFTI

USB

USBSTOR

C:\Random\FlashMProd_USB_Special_DiskMRev...

DiskMRev_GeneticProd_Mass_StorageMRev_2.00

DiskMRev_VendorCellProd_ProductCodeMRev_2.00

Hardware Profiles

Palcos

Services

ServiceState

DriverDatabase

HardwareConfig

Input

Keyboard Layout

Maps

MountedDevices

ResourceManager

ResourcePolicyStore

RNG

Select

Values: USB

Drag a column header here to group by that column

Key Name	Serial Number	ParentID Prefix	Service	Device Desc	Friendly Name	Device Name	Location Information	Installed	First Installed	Last Connected	Last Removed
VID_095F&PID_0007	76026770236051	8A1068030580	Hub	USB Input Device		VMware Virtual USB Tablet	Port_#0001.Hub_#0001	2026-06-25 13:49:...	2026-06-25 13:49:...	2026-06-25 23:44:14	
VID_095F&PID_0418	20240418000000		USBSTOR	USB Mass Storage Device		USB 2.0 Device	Port_#0001.Hub_#0001	2026-07-20 13:23:...	2026-07-20 13:23:...	2026-07-20 13:23:30	
VID_095F&PID_5678	518003114173036679		USBSTOR	USB Mass Storage Device		Disk 2.0	Port_#0001.Hub_#0001	2026-06-25 10:25:...	2026-06-25 10:25:...	2026-07-20 13:23:21	2026-07-20 13:31:24
VID_0A38&PID_0497	761be48778082	8A25ccf8b8d0	USBSTOR	USB Mass Storage Device		WCH UCI05K_V1.1	Port_#0002.Hub_#0002	2026-07-20 13:25:...	2026-07-20 13:25:...	2026-07-20 13:26:55	2026-07-20 13:27:56
ROOT_HUB	552891968b8d0		usbhub	USB Root Hub				2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:10	
ROOT_HUB0	5A36a4e5658d0		usbhub	USB Root Hub				2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:10	
ROOT_HUB30	5A1105705808d0	6A39d724f65d0	USBHUB3	USB Root Hub (USB 3.0)				2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:10	
VID_08D&PID_9230	MSFT0012345678962	76f10f4146d0	UASPStor	USB Attached SCSI (UAS) Mass Storage Device		RTL9210C	Port_#0001.Hub_#0001	2026-07-20 13:21:...	2026-07-20 13:21:...	2026-07-20 13:45:11	
VID_095F&PID_0003	6A39d724f65d0	76bdcf1c23d0	usbcomp	USB Composite Device		VMware Virtual USB Mouse	Port_#0005.Hub_#0003	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_095F&PID_0007	6A39d724f65d0	761731478d0	Hub	USB Input Device		VMware Virtual USB Tablet	Port_#0006.Hub_#0003	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_095F&PID_0002	6A39d724f65d0	76026770236051	USBHUB3	Generic USB Hub		VMware Virtual USB Hub	Port_#0007.Hub_#0003	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_095F&PID_0002	6A39d724f65d0	761be48778082	USBHUB3	Generic USB Hub		VMware Virtual USB Hub	Port_#0008.Hub_#0003	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_095F&PID_0003&M_00	76bdcf1c23d080000	8A217cb298d0	Hub	USB Input Device		VMware	000b.0000.0000.0005	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_095F&PID_0003&M_01	76bdcf1c23d080001	8A217cb298d0	Hub	USB Input Device		VMware	000b.0000.0000.0005	2026-06-25 17:12:...	2026-06-25 17:12:...	2026-07-20 13:45:12	
VID_0A38&PID_0396	3CAB724F537		WCH5820C64	USB 2.0 Ethernet Adapter	USB 2.0 Ethernet Adapter	USB 10/100 LAN	Port_#0003.Hub_#0003	2026-07-20 13:28:...	2026-07-20 13:28:...	2026-07-20 13:45:16	

Total rows: 15

Export

Type viewer

Activate Windows
Go to Settings to activate Windows.

Value: None Collapse all Hives

Hidden keys: 0

USB Detective v1.6.4 Community Edition (non-commercial use only) [usb forensic-usb 001]

File Tools View Export Help

Serial/UID	Description	First Connected (UTC)	Last Connected (UTC)	Last Disconnected (UTC)	Volume Name:Label	Drive Letter(s)	VSN	Last User
76026770236051	Flash-USB Special Disk USB Device	7/20/2026 1:25:27 PM	7/20/2026 1:26:05 PM	7/20/2026 1:27:16 PM		F:		
20240418000000	Generic Mass Storage USB Device	7/20/2026 1:23:18 PM	7/20/2026 1:23:26 PM	7/20/2026 1:23:38 PM				
518003114173036679	VendorCo ProductCode USB Device	6/26/2026 3:25:17 AM	7/20/2026 1:25:31 PM	7/20/2026 1:31:24 PM	E:\	E:		juser
012345678962	Realtek RTL9210C NVME SCSI Disk Device	7/20/2026 1:21:15 PM	7/20/2026 1:45:11 PM	7/20/2026 1:23:25 PM				
1639d76180800000	ARSENAL VIRTUAL	6/30/2026 6:50:33 PM			E:\			juser

7/20/2026 5:28:06 PM UTC: Finished processing Arsenache hive (C:\Users\juser\Documents\Target20267207150251WRL_11_V\VF\Windows\AppCompat\Programs\Arsenache hive).

7/20/2026 5:28:06 PM UTC: No event logs provided.

7/20/2026 5:28:06 PM UTC: Performing additional correlation across provided artifacts.

7/20/2026 5:28:07 PM UTC: Populating results grid and checking for duplicate timestamps.

7/20/2026 5:28:07 PM UTC: 5 devices were identified by USB Detective!

7/20/2026 5:28:07 PM UTC: Processing of WRL_11_V is complete.

Activate Windows
Go to Settings to activate Windows.

Trustworthy Consistency Levels

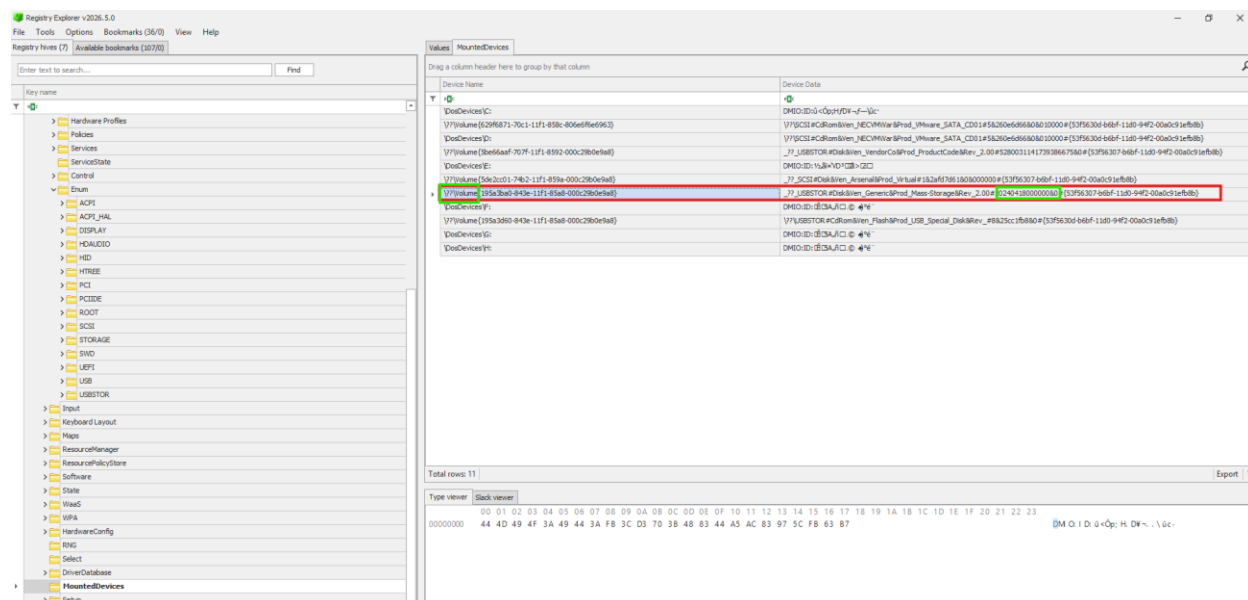
Not Calculated Low Medium High

Low High

It was noted during this examination that the serial number can also be identified from the numbers assigned to the drive in the folder name listed under USBSTOR, which may be either globally or system unique, and that the last write time can be identified in the last write time field associated with the serial number in that folder name.

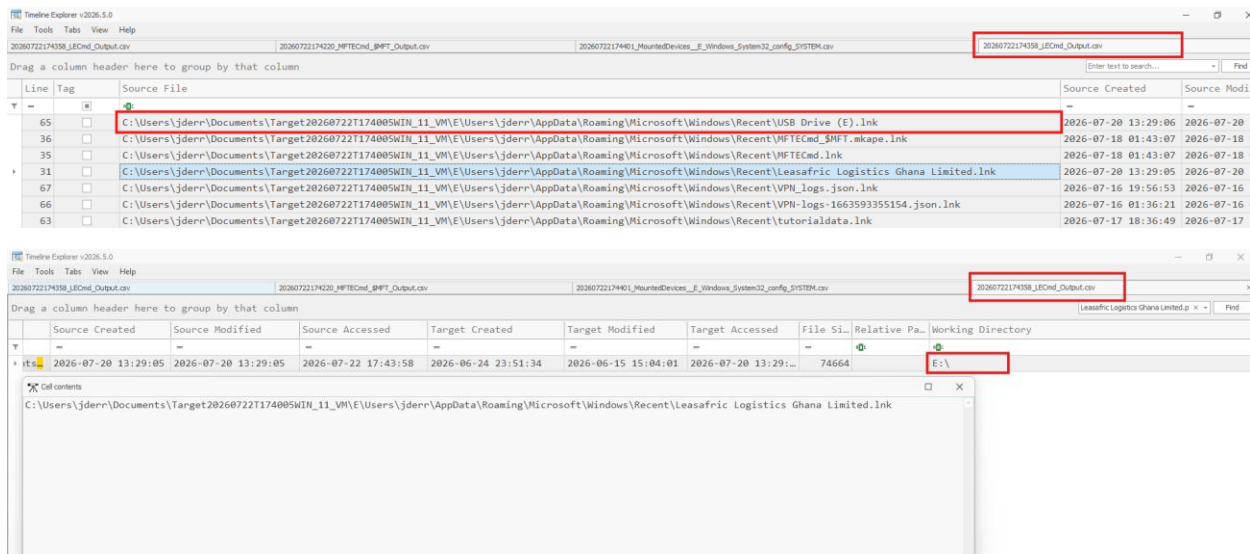
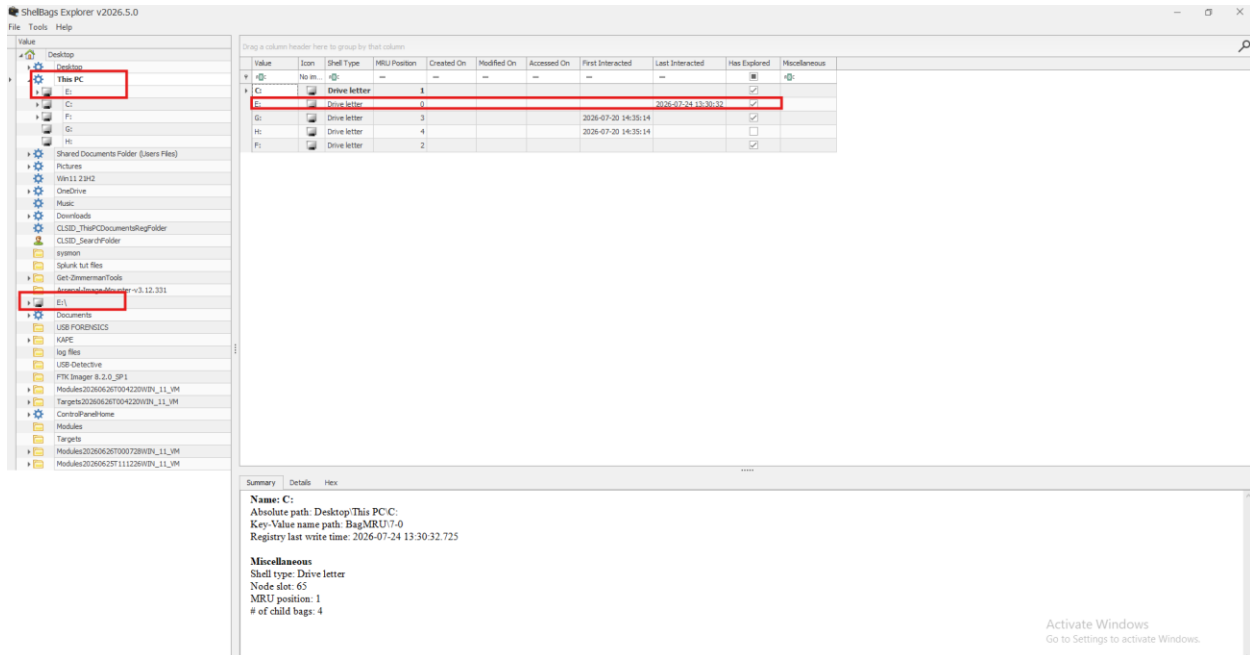
Assigned Drive Letter

MountedDevices was checked to determine whether the drive had mounted properly, since a drive letter could not initially be identified elsewhere. The serial number was present, confirming this was the drive under examination, however no drive letter was assigned to it, only a \??\Volume reference.



Possible reasons for this include the use of a low-quality flash controller in the device, which can cause a disconnect during initialization and interrupt Windows partway through mounting, corrupted partition tables, or automount being disabled. Given that other drives on the system had letters successfully assigned, the low-quality flash controller explanation was assessed as the most likely cause.

The drive was later confirmed to have been assigned drive letter E, through LNK file and ShellBags artifacts. The date and time recorded in ShellBags aligned with when the USB device was first installed, and the Has Explored box was checked for this entry.



USB Device Registry Artifacts

Attribute	Value
USB Device Name	Generic Mass Storage USB Device
Manufacturer	MASS STORAGE USB DEVICE / Ven Generic
Model	VID_349C&PID_0418
Serial Number	20240418000000
Installed	2026 07 20 13:23:17.4716162
First Installed	2026 07 20 13:23:17.4716162

Last Connected	2026 07 20 13:23:17.4639635
Last Removed	2026 07 20 13:23:30.1536493
Port Number	Port_#0001.Hub_#0005
User Profile	Jderr
Timezone	UTC
Assigned Letter	E (confirmed via LNK file and shellbags)

The manufacturer field appears as Generic MASS STORAGE USB DEVICE and Ven Generic because the device is unbranded.

Files Opened from the USB Device

The LECmd.csv output was reviewed, filtered by Drive Type to Removable storage media (Floppy, USB). The local path and working directory values both began with E:\. A file named Leasafric Logistics Ghana Limited.pdf, opened from the USB device, was identified in this output. The source created and accessed timestamps for this file matched the timeline of when the USB device was inserted into the system. This file was also visible under the Target Name field in NTUSER.DAT.csv, along with an entry for USB DRIVE (E:), with ascending MRU positions and corresponding timestamps.

Timeline Explorer v2026.5.0												
File Tools Tabs View Help												
20260722174401_RacoonEvoics_8_users_jderr_NTUSER.DAT.csv				20260722174354_AutomaticDestinations.csv				20260722174358_LECmd_Output.csv				
Drag a column header here to group by that column												
	File...	H...	Volume Serial Number	Drive Type	V...	Ne...	Local Path	Arg...	Target ID	Absolute Path	Target MFT Entry Num...	Target
HFTECmd	File...	H...	7A789FD3	Removable storage media (Floppy, USB)			E:\		My Computer\E:			
	File...	H...	7A789FD3	Removable storage media (Floppy, USB)			E:\Get-ZimmermanTools\KAPE\Modules\EZTool...		My Computer\E:\	0x5D8	0x1	
	File...	H...	7A789FD3	Removable storage media (Floppy, USB)			E:\Get-ZimmermanTools\KAPE\Modules\EZTool...		My Computer\E:\	0x5CD	0x1	
	File...	H...	7A789FD3	Removable storage media (Floppy, USB)			E:\Leasafric Logistics Ghana Limited.pdf		My Computer\E:\	0x984	0x1	
er\sysmon-config-master	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\VPN_logs...		Desktop\	0x59D	0x6	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\VPN_logs...		Desktop\	0x59D	0x6	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Splunk tu...		Desktop\	0x27A1F	0x7	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Documents\Timesto...		Documents\	0xA7F	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Targets		My Computer\C:\Users\De...	0x19A3E	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\Downloads\sysmon-config-ma...		Downloads\	0x19A3E	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\sysmon		Downloads\	0x26D34	0x5	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\Downloads\sysmon-config-ma...		Downloads\	0x19A3A	0x3	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\Downloads\sysmon		Downloads\	0x11253	0x1	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Splunk tu...		Desktop\	0xB3CB	0x4	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		My Computer\C:\Users\PI...	0x2585E	0x7	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		My Computer\C:\Users\PI...	0x0		
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xAA7	0x0	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		My Computer\C:\Users\PI...	0xC06	0x3	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xB21	0x4	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xB2F	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xB27	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xAD6	0x3	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xAD2	0x4	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Pictures\Screensh...		Pictures\Screenshots\	0xA9C	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Modules20...		Desktop\	0x1946D	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Nina_Morg...			0x26253	0x5	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\New Folder			0xAEB	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Modules			0xAED	0x2	
	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Documents\Logisti...		Documents\	0x26488	0x5	
File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Documents\Logisti...		Documents\	0x26487	0x5		
File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Leasafric...			0x26249	0x5		
0626T004220WIN_11_VMF...	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Modules20...		Desktop\	0x194CC	0x2	
0626T004220WIN_11_VMF...	File...	H...	C4806E0F	Fixed storage media (Hard drive)			C:\Users\jderr\OneDrive\Desktop\Modules20...		Desktop\	0x194CB	0x2	

File Copying Between USB and Computer

No evidence was found of files being copied between the USB device and the computer. Both the local path and the TargetID Absolute Path fields reviewed show only the E:\ path.

Registry Hive State

Registry Explorer reported that the SYSTEM hive was in a dirty state, with associated transaction logs (.LOG1 and .LOG2) present and requiring replay. This indicates the hive did not necessarily represent the most recently committed state at the time of acquisition.

This condition was anticipated, since the evidence was acquired from a live Windows system, and registry hives on live systems are commonly left in a dirty state because pending transactions have not yet been committed to the hive files. The presence of transaction logs is therefore considered an expected characteristic of live acquisitions rather than an indication of evidence corruption.

As a result, USB related artifacts, including connection history, registry key LastWrite timestamps, and derived timestamps such as Installed, First Installed, Last Connected, and Removed, may be incomplete, outdated, or may reflect only the last committed transaction. These artifacts were interpreted with appropriate caution during this examination.

A separate technical guide was previously produced documenting the causes of dirty registry hives, their impact on forensic analysis, and the process of replaying registry transaction logs to recover the most current registry state. That methodology applies directly to this case and provides additional context for interpreting the registry evidence.

Reference: Registry Transaction Log Replay (Dirty Hive Recovery), [insert link to documentation](#).

Additionally, the SYSTEM hive contained the registry value NtfsDisableLastAccessUpdate, with a value of 2147483650 (0x80000002). This indicates that NTFS last access timestamp updates were disabled under a system managed configuration. As a result, NTFS Last Access timestamps may not reliably reflect file access activity on this system.

Timeline of Events

Date and Time	Event
6/25/2026	Windows 11 Pro installed on host system
2026 07 20, 13:23:17.4639635	USB device (Generic Mass Storage USB Device, Serial Number 20240418000000) last connected
2026 07 20, 13:23:17.4716162	USB device installed and first installed
2026 07 20, 13:23:30.1536493	USB device removed
7/20/2026	File Leasafric Logistics Ghana Limited.pdf opened from USB device (E:) using Microsoft Edge (Chromium), under user profile Jderr
7/20/2026	E01 image (NFTS_USB_FORENSICS.E01) acquired using FTK Imager v8.2.0.33 SP1

Conclusion

The examination identified a Generic Mass Storage USB Device, Serial Number 20240418000000, connected to the win_11_vm system on 2026 07 20 at 13:23:17 and removed at 13:23:30. The device was assigned drive letter E, confirmed through LNK file and ShellBags artifacts, although this letter was not visible in MountedDevices due to no letter being assigned there. The user profile Jderr interacted with the device, and Microsoft Edge (Chromium) was used to open the file Leasafric Logistics Ghana Limited.pdf from the device. No evidence was found that files were copied between the USB device and the host computer. Registry hive dirty state and the disabling of NTFS last access timestamp updates were both noted as factors affecting the reliability and completeness of certain timestamp-based artifacts, and findings were interpreted accordingly.

Exhibits and Appendices

- Host system details (see Host Details table above)
- E01 image hash values, MD5 and SHA1 (see Evidence and Chain of Custody table above)
- SYSTEM\CurrentControlSet\Enum\USBSTOR registry key
- SYSTEM\CurrentControlSet\Enum\USB registry key
- MountedDevices registry key
- SetupAPI.dev.log
- LECmd.csv
- NTUSER.DAT.csv
- AutomaticDestinations.csv
- FileFolderAccess, Jump Lists CSV output
- Registry Transaction Log Replay (Dirty Hive Recovery) technical guide, [insert link to documentation](#)